JAY BOMBATKAR

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Embedded Systems Engineer | PCB Design | Firmware Development

PROFESSIONAL SUMMARY

Electronics and Telecommunication Engineering final-year student (SGPA: 8.85) with hands-on experience in embedded systems design, PCB layout, and microcontroller firmware development. Proficient in STM32, PIC microcontrollers, communication protocols (UART, SPI, I2C, CAN Bus), and PCB design tools (KiCad, Altium Designer). Experienced in hardware validation, sensor testing, and debugging with oscilloscopes. Seeking an entry-level role in Embedded Systems Engineering, Hardware Design, or Electronics R&D.

TECHNICAL SKILLS

Programming Languages:
Embedded C, Python, SQL
Microcontrollers & RTOS:
STM32, PIC, FreeRTOS, Interrupts, Timer Programming
Communication Protocols:
UART, SPI, I2C, USB, CAN Bus, RS-485
PCB Design Tools:
Altium Designer, KiCad, LTspice
Debugging & Test Equipment:
Oscilloscope, Multimeter, Logic Analyzer
Development Tools:
MPLAB X IDE, MPLAB IPE, Git, GitHub, Linux

WORK EXPERIENCE

Embedded Systems & Hardware Test Intern | SuryaLogix Pvt. Ltd. | Feb 2026 - Apr 2026

- Performed functional and performance testing of environmental and solar sensors
- Conducted PCB-level debugging and fault isolation using oscilloscope and multimeter
- Verified analog and digital sensor outputs (RS-485, 4-20mA) across varied conditions
- Executed production testing and quality assurance procedures per industry standards
- Collaborated on root cause analysis, issue resolution, and process improvement

PROJECTS

CNC Controller Using MPG | Ongoing

Developed microcontroller-based CNC jogging controller with MPG rotary encoder for precise manual control. Designed 2-layer PCB in KiCad with signal routing and EMI/EMC mitigation. Implemented Embedded C firmware for quadrature encoding and real-time axis control. Performed hardware testing using oscilloscope.

STM3

Designed production-ready STM32 PCB using Altium Designer with 4-layer layout and DRC. Implemented power supply circuitry, USB/UART interfaces. Performed component placement optimization and impedance-controlled routing. Generated complete Gerber files, NC drill files, and BOM.

EDUCATION

Bachelor of Engineering - Electronics and Telecommunication | 2022 - 2026

Savitribai Phule Pune University | SGPA: 8.85 / 10.0

CERTIFICATIONS

- PCB Design with KiCad - Udemy
- Hardware Design using PIC Microcontroller - Udemy

ACHIEVEMENTS & LEADERSHIP

- Project Lead at Project Expo 2026, PVPIT Bavdhan
- Event Lead at Quiz Competition 2024, PVPIT Bavdhan